

BIOE 530: Biomaterials Science and Regenerative Scaffolds

Term: Fall 2026 | Instructor: Dr. Robert Langer (rlanger@bioeng.univ.edu)

Course Description & Overview:

Biological host responses, polymer degradation mechanisms, 3D bioprinting technologies, and regulatory approval for medical implants.

COURSE ASSIGNMENTS & WEIGHT DISTRIBUTION

Assignment Description	Weight	Due Module	Deliverable Format
Polymeric Biomaterial Characterization Lab	25%	Week 04	Lab Report & Rheology Data
Tissue Engineering Bioreactor Design	30%	Week 08	CAD Specs & Perfusion Analysis
FDA 510(k) Medical Device Regulatory Dossier	35%	Week 11	Regulatory Submission Package
Biosafety & Bioethics Quizzes	10%	Week 06	Proctored Quizzes

WEEKLY TERM SCHEDULE & READING REQUIREMENTS

Timeline	Technical Topic Focus	Assigned Readings & Docs
Week 01	Module 01: Protein Adsorption & Biocompatibility	Ratner et al. (Ch. 1–3)
Week 02	Module 02: Biodegradable Synthetic Polymers (PLGA)	Langer (1990); Ratner (Ch. 5)
Week 03	Module 03: Extracellular Matrix Scaffolds	Badylak et al. (2009)
Week 04	Module 04: Scaffold Fabrication & 3D Bioprinting	Murphy & Atala (2014)
Week 05	Module 05: Cellular Mechanotransduction Dynamics	Discher et al. (2005)
Week 06	Module 06: Foreign Body Reaction & Macrophages	Anderson et al. (2008)
Week 07	Module 07: Vascularization of Engineered Tissue	Jain et al. (2005)
Week 08	Module 08: Orthopedic Implants & Titanium Alloys	Hench (1991); ASTM F136 Standards
Week 09	Module 09: Hydrogels for Controlled Drug Release	Peppas et al. (2000)
Week 10	Module 10: Regulatory Pathways & ISO 10993	FDA Device Guidance Specs
Week 11	Module 11: Clinical Translation Symposium	Translation Whitepapers